

Multiset Dimension of Cylindrical Graphs: An Infinite Family and a Certified Census

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Abstract

Multiset dimension asks how few landmarks distinguish every graph vertex when only the unordered multiset of landmark distances is observed. For the cylindrical graph $P_m \square C_{6m}$ and every even integer $m \geq 2$, we prove that the vertices $(0, 0)$, $(0, 2m)$, and $(m - 1, 0)$ form a multiset basis of $P_m \square C_{6m}$; consequently, $\dim_m(P_m \square C_{6m}) = 3$. The proof recovers the labelled distance data from its multiset by parity and then rules out the only possible label swap using a four-piece description of the relevant cycle-distance locus. This gives an exact infinite family. Complementing it, a separately implemented exhaustive checker determines the maximum path order resolved by any three boundary-row landmarks for every cycle order $4 \leq n \leq 100$, and gives the complete multiset-dimension table for $3 \leq m \leq 7$ and $3 \leq n \leq 14$. It closes all 55 entries in the $4 \leq n \leq 14$ subbox that were not previously exact, identifies exactly nine dimension-five cylinders, and improves six published five-landmark constructions to four landmarks. Explicit witnesses, exhaustive lower-bound searches, and strict-schema corruption tests accompany the classifications; the computation is independent of the proof of the infinite family.

Keywords: multiset dimension; resolving set; Cartesian product; cylindrical graph; certified computation.

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1 Introduction

Let G be a connected graph and $W \subseteq V(G)$. The *multiset code* of a vertex v with respect to W is

$$r_m(v \mid W) = \{d(v, w) : w \in W\},$$

where multiplicities are retained. The set W is *multiset resolving* if distinct vertices have distinct codes. The minimum cardinality of such a set is the multiset dimension $\dim_m(G)$; if no such set exists, then $\dim_m(G) = \infty$.

The multiset-resolving notion was introduced by Saenpholphat and independently rediscovered by Simanjuntak, Siagian, and Vetrík [9, 10]. The latter paper proved that no connected graph has multiset dimension two and that a connected graph has dimension one precisely when it is a path [3, 9, 10]. The equivalence between multiset resolving sets and ID-colorings was later made explicit by Hakanen and Yero [5]. For broader context, see the recent surveys [1, 4] and the structural work of Bong and Lin [2].

This paper sharpens what is known for the cylindrical product $P_m \square C_n$. Marcelo, Tolentino, Garciano, and Buot proved that $\dim_m(P_m \square C_n) = 3$ whenever $m \geq 2$ and $n \geq 8m + 1$, using three landmarks in one boundary row [8]. The gap between that sufficient region and the small exact cases leaves the natural question of whether the threshold can be brought closer to $6m$.

Our main theorem answers this question on the infinite diagonal $n = 6m$ for every even m . Two certified finite results accompany the proof. First, we optimize all three-landmark boundary-row patterns through $n = 100$. Second, we determine every exact value in the rectangle $3 \leq m \leq 7$, $3 \leq n \leq 14$. The computation is not used to prove the infinite theorem; it instead delineates the finite landscape and supports a precisely scoped conjecture. The accompanying artifact contains the raw records, a separately implemented verifier, corruption tests, and a small Lean companion.

2 Preliminaries and known cylinder results

We write P_m for the path on rows $0, \dots, m-1$ and C_n for the cycle on $\mathbb{Z}_n$. Distances in their Cartesian product satisfy

$$d((i, j), (r, s)) = |i - r| + \min\{|j - s|, n - |j - s|\}. \quad (1)$$

The cylinder paper cited above also records the previously determined special rows: for $m \geq 3$, $\dim_m(P_m \square C_3) = 3$ with basis $\{(0, 0), (0, 1), (2, 0)\}$; the underlying cycle result is $\dim_m(C_n) = 3$ for $n \geq 6$, with $\{0, 1, 3\}$ resolving [10, Theorem 4.1]. For prisms $P_2 \square C_n$ the dimension is three for $n \geq 11$, $n \neq 13$, four for $n \in \{8, 9, 10, 13\}$, five for $n \in \{6, 7\}$, and infinite for $n < 6$ [6, 7, 8].

Marcelo et al. also give four-landmark upper bounds for all $m \geq 6$, $n \geq 3$ and for all $m \geq 1$, $n \geq 12$, and prove finiteness for every cylinder in the range $m, n \geq 3$ [8, Theorems 3.3, 3.6, and 3.10]. Their abstract states $m \geq 3$ for the $n \geq 8m + 1$ dimension-three result, whereas the displayed theorem gives $m \geq 2$; throughout we follow the theorem statement.

Lemma 1 (Cycle distance-pair locus). *For $0 \leq j < 6m$, put $a = d_{C_{6m}}(j, 0)$ and $b = d_{C_{6m}}(j, 2m)$. Then*

$$(a, b) = \begin{cases} (j, 2m - j), & 0 \leq j \leq 2m, \\ (j, j - 2m), & 2m \leq j \leq 3m, \\ (6m - j, j - 2m), & 3m \leq j \leq 5m, \\ (6m - j, 8m - j), & 5m \leq j < 6m. \end{cases} \quad (5)$$

The pair (a, b) determines j modulo $6m$.

Proof. The four expressions follow by comparing the two arcs to columns 0 and $2m$. They lie, respectively, on the lines $a + b = 2m$, $a - b = 2m$, $a + b = 4m$, and $b - a = 2m$, and each parametrization is injective on its stated interval. The six pairwise line relations are

pieces	intersection	parameters
1, 2	$(2m, 0)$	$j = 2m$ in both
1, 3	$\emptyset$	parallel
1, 4	$(0, 2m)$	$j = 0$ and $j = 6m$
2, 3	$(3m, m)$	$j = 3m$ in both
2, 4	$\emptyset$	parallel
3, 4	$(m, 3m)$	$j = 5m$ in both.

The fourth interval is half-open, so $j = 6m$ is excluded; in any case it is the same cycle column as $j = 0$. Thus every overlap is consistent with a unique residue class modulo $6m$. $\square$

3 An exact infinite family

Theorem 2. *For every even integer $m \geq 2$,*

$$\dim_m(P_m \square C_{6m}) = 3.$$

More precisely,

$$W_m = \{(0, 0), (0, 2m), (m-1, 0)\}$$

is a multiset basis.

Proof. Put $N = 6m$. For a vertex $v_t = (i_t, j_t)$, $t \in \{1, 2\}$, write

$$a_t = d_{C_N}(j_t, 0), \quad b_t = d_{C_N}(j_t, 2m),$$

and label its three distances to W_m by

$$X_t = i_t + a_t, \quad Y_t = i_t + b_t, \quad Z_t = m - 1 - i_t + a_t. \quad (2)$$

On an even cycle, the two complementary arc lengths between any two columns have the same parity; hence cycle distance has the parity of the corresponding column displacement. Because the landmark columns differ by the even quantity $2m$, we have $a_t \equiv b_t \pmod{2}$, so $X_t \equiv Y_t \pmod{2}$. In contrast,

$$Z_t - X_t = m - 1 - 2i_t$$

is odd because m is even. Thus the unordered code identifies Z_t as its unique entry of one parity and identifies $\{X_t, Y_t\}$ as the two entries of the other parity.

For either possible ordering of the two same-parity entries, the ordered quantities would be reconstructed by

$$i = \frac{m - 1 - Z + X}{2}, \quad a = X - i, \quad b = Y - i. \quad (3)$$

Suppose that two vertices v_1 and v_2 receive the same code. Then $Z_1 = Z_2$ and $\{X_1, Y_1\} = \{X_2, Y_2\}$. If $X_1 = X_2$ and $Y_1 = Y_2$, then (3) gives $i_1 = i_2$, $a_1 = a_2$, and $b_1 = b_2$; Lemma 1 then gives $j_1 = j_2$. Otherwise we may assume $X_1 = Y_2$, $Y_1 = X_2$, and $X_1 \neq Y_1$. Solving the two swap equalities together with $Z_1 = Z_2$ gives

$$i_2 = i_1 + \frac{b_1 - a_1}{2}, \quad a_2 = \frac{a_1 + b_1}{2}, \quad b_2 = \frac{3a_1 - b_1}{2}. \quad (4)$$

Since $a_1 \equiv b_1 \pmod{2}$, all three right-hand sides in (4) are integers. We show that no genuine second vertex has these parameters, using the four pieces of Lemma 1. On the second piece, $b_1 - a_1 = -2m$, so $i_2 = i_1 - m < 0$. On the fourth, $b_1 - a_1 = 2m$, so $i_2 = i_1 + m \geq m$.

On the first piece, $a_1 + b_1 = 2m$. Equation (4) gives $a_2 = m$, $b_2 = 2a_1 - m$, and $i_2 = i_1 + m - a_1$. A cycle vertex at distance m from column 0 is column m or $5m$, whose distance from column $2m$ is respectively m or $3m$. Hence a valid pair with first coordinate m is only (m, m) or $(m, 3m)$. The first forces $a_1 = b_1$, contrary to $X_1 \neq Y_1$; the second forces $a_1 = 2m$, whence $i_2 = i_1 - m < 0$.

On the third piece, $a_1 + b_1 = 4m$. Equation (4) gives $a_2 = 2m$, $b_2 = 2a_1 - 2m$, and $i_2 = i_1 + 2m - a_1$. A cycle vertex at distance $2m$ from column 0 is column $2m$ or $4m$, giving the pair $(2m, 0)$ or $(2m, 2m)$. The first forces $a_1 = m$ and then $i_2 = i_1 + m \geq m$; the second forces $a_1 = b_1$, again contrary to $X_1 \neq Y_1$. Therefore no distinct second vertex can realize the swapped assignment.

The swapped assignment is therefore impossible, while the non-swapped assignment is unique by Lemma 1. The three-landmark set W_m resolves the cylinder.

The graph is connected and is not a path, so the general lower bound $\dim_m \geq 3$ applies [10]. Thus W_m is a basis. $\square$

Remark 3 (Why evenness appears). For odd $m \geq 3$, the same displayed triple is not resolving. The vertices

$$\left(0, \frac{m+3}{2}\right), \quad \left(1, \frac{m+1}{2}\right)$$

both receive the code

$$\left\{\frac{m+3}{2}, \frac{3m-3}{2}, \frac{3m+1}{2}\right\}.$$

This only shows that this basis fails for odd m ; it does not imply that the multiset dimension exceeds three.

Remark 4 (The prism endpoint). The case $m = 2$ is $P_2 \square C_{12}$ and is already covered by the exact prism results cited above. The uniform proof is retained because it exposes the all-even mechanism; the portion beyond the known prism endpoint begins at even $m \geq 4$.

4 Certified boundary-row optimization

Let a three-landmark set lie in row zero at columns $S = \{w_1, w_2, w_3\} \subseteq \mathbb{Z}_n$, and define

$$c_S(j) = \text{sort}(d_{C_n}(j, w_1), d_{C_n}(j, w_2), d_{C_n}(j, w_3)).$$

For a vertex (i, j) ,

$$r_m((i, j) \mid W) = c_S(j) + i(1, 1, 1). \quad (6)$$

Thus S resolves $P_m \square C_n$ exactly when the cycle codes are distinct and no two differ by $t(1, 1, 1)$ for an integer $1 \leq |t| \leq m-1$. Define $\text{cap}_n(S)$ to be the largest such path order m , with value 0 if S does not resolve C_n and value ∞ if it resolves every path order, and put

$$\text{cap}(n) = \max_{|S|=3} \text{cap}_n(S).$$

Lemma 5 (Capacity semantics). *Suppose the cycle codes $c_S(j)$ are distinct. Let g be the smallest positive integer such that two of them differ by $g(1, 1, 1)$; put $g = \infty$ if no such pair exists. Then S resolves $P_m \square C_n$ exactly for $1 \leq m \leq g$. Hence $\text{cap}_n(S) = g$.*

Proof. Equation (6) shows that vertices in the same row collide only when their cycle codes agree, and vertices in rows differing by $t \neq 0$ collide exactly when their cycle codes differ by $t(1, 1, 1)$. Such a collision occurs among the rows $0, \dots, m-1$ precisely when $1 \leq |t| \leq m-1$. The first positive shift g therefore causes failure at path order $g+1$, but not before. $\square$

Proposition 6 (Certified capacity formula). *For $4 \leq n \leq 100$, write $n = 6k + r$, $0 \leq r \leq 5$. Then*

$$\text{cap}(n) = \begin{cases} 1, & n = 8, \\ k, & r \in \{3, 4\}, \\ k, & r \in \{0, 1\} \text{ and } k \text{ is odd}, \\ k-1, & r \in \{0, 1\} \text{ and } k \text{ is even}, \\ k, & r = 2 \text{ and } k \text{ is even}, \\ k-1, & r = 2 \text{ and } k \text{ is odd, } n \neq 8, \\ k+1, & r = 5 \text{ and } k \text{ is odd}, \\ k, & r = 5 \text{ and } k \text{ is even}. \end{cases} \quad (7)$$

In particular, the formula gives $\text{cap}(4) = \text{cap}(5) = 0$. For $6 \leq n \leq 100$, the same value is equivalently the largest integer $m \geq 1$ satisfying: $n \geq 6$ when $m = 1$; $n \geq 6m$ and $n \neq 6m + 2$ when $m \geq 2$ is odd; or $n \geq 6m - 1$ and $n \notin \{6m, 6m + 1\}$ when m is even.

Exact computational proof. By cycle rotation, every three-subset of columns has a representative $\{0, a, b\}$ with $0 < a < b < n$. The generator and verifier enumerate every anchored triple. The certificate stores, for each n , one attaining triple and the certified optimum. For every anchored triple, the verifier computes all sorted cycle codes, rejects duplicate codes, and finds the smallest positive constant shift between two codes; Lemma 5 identifies this shift with the height capacity. The certificate contains 98 rows for $3 \leq n \leq 100$; the proposition concerns the 97 orders $4 \leq n \leq 100$, where the recomputed maximum agrees with (7). Through $n = 45$, a separate direct pairwise implementation cross-checks every anchored triple; all 14,190 anchored triples agree. This count includes the $n = 3$ certificate row; restricting to $4 \leq n \leq 45$ gives 14,189. $\square$

Corollary 7. *For every $6 \leq n \leq 100$ and every $1 \leq m \leq \text{cap}(n)$,*

$$\dim_m(P_m \square C_n) = 3.$$

In particular,

$$\dim_m(P_{16} \square C_{95}) = 3,$$

witnessed by boundary columns $\{0, 16, 47\}$, since $\text{cap}(95) = 16$.

Proof. A triple attaining capacity at least m is a three-landmark resolving set by (6). The graph is connected and not a path, so the lower bound three applies. For the displayed example, the certificate records the stated columns. The previous general sufficient condition $n \geq 8m + 1$ would require $n \geq 129$ when $m = 16$ [8]. $\square$

Remark 8 (The cross-row point). For even m , Proposition 6 excludes $n = 6m$ from the boundary-row criterion, but Theorem 2 supplies a cross-row triple there. For example, $\text{cap}(24) = 3 < 4$, so no boundary-row triple resolves $P_4 \square C_{24}$, whereas W_4 does. This is the structural distinction between the infinite theorem and the boundary computation.

5 A complete small exact table

Marcelo et al. already supplied four-landmark resolving sets whenever $m \geq 6$ or $n \geq 12$, and explicit sets of cardinality four or five for the remaining pairs $3 \leq m \leq 5$, $4 \leq n \leq 11$ [8, Theorems 3.3, 3.4, 3.6 and Table 1]. Thus the five $n = 3$ entries below were previously exact, and among the other 55 entries, 49 inherit an upper bound of the displayed cardinality. The six strict improvements from five landmarks to four occur at $(3, 9)$, $(4, 7)$, $(4, 9)$, $(5, 4)$, $(5, 5)$, $(5, 9)$. The principal new content is the exhaustive optimality proof for all 55 previously non-exact entries.

Theorem 9. *For $3 \leq m \leq 7$ and $3 \leq n \leq 14$, the exact values of $\dim_m(P_m \square C_n)$ are as follows.*

$m \backslash n$	3	4	5	6	7	8	9	10	11	12	13	14
3	3	5	5	5	5	4	4	4	4	4	4	4
4	3	5	5	5	4	4	4	4	4	4	4	4
5	3	4	4	5	5	4	4	4	4	4	4	4
6	3	4	4	4	4	4	4	4	4	4	4	4
7	3	4	4	4	4	4	4	4	4	4	4	4

Consequently, the nine bold entries are precisely the dimension-five cylinders in this box.

Exact computational proof. For every one of the 60 parameter pairs, the certificate supplies an explicit resolving set of the displayed size. For each entry equal to four, the verifier directly enumerates every three-subset of the vertex set, without symmetry reduction, and rejects all of

them. For each entry equal to five, it directly enumerates every set of sizes one through four and rejects all of them. For the dimension-four entries, the general facts that a connected non-path has neither dimension one nor dimension two exclude sizes one and two; the checker also enumerates those sizes as a redundant semantic test. The five dimension-three entries have explicit witnesses and the same general lower bound. The strict-schema verifier checks every row without embedding the expected table as an answer key. $\square$

Remark 10 (Nonmonotonicity). The table is not monotone in the cycle order: for example,

$$\begin{aligned}\dim_m(P_5 \square C_4) &= \dim_m(P_5 \square C_5) = 4, \\ \dim_m(P_5 \square C_6) &= \dim_m(P_5 \square C_7) = 5, \quad \dim_m(P_5 \square C_8) = 4.\end{aligned}$$

Separately, the lower-bound search tests landmark-set sizes independently; appending a landmark to a multiset resolving set need not preserve resolution because the landmark labels are forgotten.

6 A candidate formula

The capacity formula concerns only boundary-row triples and therefore does not establish lower bounds against arbitrary landmarks. Together with the complete small table and Theorem 2, it nevertheless suggests the following complete conjectural characterization. It incorporates known cases and extends into the unresolved regime.

Conjecture 11. *Let $m \geq 3$ and $n \geq 4$. Then $\dim_m(P_m \square C_n) = 3$ precisely when*

$$\begin{cases} n \geq 6m \text{ and } n \neq 6m + 2, & m \text{ odd}, \\ n \geq 6m - 1 \text{ and } n \neq 6m + 1, & m \text{ even}. \end{cases}$$

The restriction is essential. When $m = 1$, the graph is a cycle; for example C_8 has dimension three with resolving set $\{0, 1, 3\}$, so the odd- m formula must not be extrapolated to $m = 1$. When $m = 2$, the complete prism row is already known and agrees exactly with the displayed even criterion; the point $n = 12$ is its non-boundary instance, analogous to Theorem 2. When $n = 3$, the published basis $\{(0, 0), (0, 1), (2, 0)\}$ gives dimension three for every $m \geq 3$, while the $m = 1, 2$ cases have infinite dimension [6, 7, 8]. Theorem 2 proves the non-boundary point $n = 6m$ for every even m . Two independent tasks remain. First, one must construct three-landmark resolving sets for every pair in the asserted positive region not covered by Theorem 2, Proposition 6, or prior results. Second, one must prove nonexistence of arbitrary three-landmark resolving sets below the proposed threshold and at the isolated exceptional orders. The conjecture is consistent with the complete small table, the certified boundary-row capacities through $n = 100$, and finite arbitrary-landmark frontier searches; none of these supplies either missing infinite argument.

7 Verification, scope, and reproducibility

The supplementary artifact contains raw JSON records, a dependency-free Python verifier, and seventeen negative tests that deliberately corrupt the certificate pathways. The finite lower bounds in Theorem 9 are checked by an unreduced enumeration that is algorithmically distinct from the symmetry-reduced discovery search. The capacity checker reconstructs every record and performs the direct 14,190-triple cross-check described in Proposition 6. A complete run additionally checks all 1,455 instances of the closed-form boundary criterion for $2 \leq m \leq 16$ and all fifteen supplied even-family witnesses through $m = 30$.

An arithmetic sanity-check file proves unordered-pair symmetry and three rational identities used after the swap substitution. The pinned Lean 4.19.0 project builds against the exact mathlib dependency recorded by its lock file. It proves no parity, integrality, cycle-distance, resolving-set, or graph-theoretic statement and is not evidence for the main theorem; the Python certificates establish the finite exhaustive claims.

The exact capacity formula is certified only through $n = 100$, and the small classification only in the displayed box. Neither computation proves Conjecture 11. A dated search record in the artifact reports that no prior occurrence of the all-even basis or the $6m$ family was found through 14 August 2026 UTC; this is evidence of diligence, not an absolute priority claim. The already known $m = 2$ instance is expressly excluded from the portion of the family advanced beyond the prism literature.

Data and software availability

The preprint, editable source, machine-readable certificates, exact verification software, negative tests, and scoped Lean companion are available in the versioned Zenodo record at <https://doi.org/10.5281/zenodo.21925458>. The deposit includes SHA-256 manifests and separates the offline Python certificate from the narrower Lean arithmetic checks.

Declarations

Conflicts of interest. The author declares no competing interests.

Author contributions. The named author conceived and directed the project, set its scope and research questions, guided the literature searches and pre-submission sweeps, selected the computational and formal directions to pursue, made the final decisions about which system-generated analyses and prose to retain or revise, and approved the manuscript.

Use of generative AI. The author used OpenAI Codex and Anthropic Claude (accessed August 2026) for literature-search assistance, automated analysis and drafting, programming, auditing, and language editing. Claude additionally provided a secondary automated adversarial audit, not independent peer review. Their outputs were non-authoritative suggestions. The named author directed the project and literature sweeps, chose which analyses and prose to retain or revise, reviewed the assembled arguments, code, citations, and manuscript, and approved the release. Neither system is listed as an author.

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